

Task 10: Firewall Configuration & Testing

1) Understand Firewall Concepts:

- A firewall controls incoming and outgoing network traffic
- It enforces security rules based on ports, IPs, and protocols
- Firewalls help reduce attack surface
- Examples: UFW, Windows Firewall, iptables

2) Check Firewall Status:

```
File Actions Edit View Help
(hari@vbox)-[~]
$ sudo ufw status verbose
[sudo] password for hari:
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
```

Firewall status was checked to understand existing security posture.

3) Set Default Rules:

```
(hari@vbox)-[~]
$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)

(hari@vbox)-[~]
$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
```

Default deny policy was applied to minimize unauthorized access.

4) Allow Essential Ports:

Allow SSH:

```
(hari@vbox)-[~]
$ sudo ufw allow ssh
Skipping adding existing rule
Skipping adding existing rule (v6)
```

Allow HTTP (Apache / DVWA):

```
(hari@vbox)-[~]
$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
```

Allow HTTPS:

```
(hari@vbox)-[~]
$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
```

Check rules:

```
(hari@vbox)-[~]
$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 443/tcp ALLOW IN Anywhere
[ 4] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 5] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 6] 443/tcp (v6) ALLOW IN Anywhere (v6)
```

5) Deny / Block a Port:

Block FTP port:

```
(hari@vbox)-[~]  
$ sudo ufw deny 21/tcp  
Rule added  
Rule added (v6)
```

Test:

```
(hari@vbox)-[~]  
$ nmap -p 21 localhost  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-05 00:44 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0018s latency).  
Other addresses for localhost (not scanned): ::1  
  
PORT      STATE SERVICE  
21/tcp    closed ftp  
  
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

Blocking unused ports reduces exposure to attacks.

6) Test Connectivity:

Test allowed port:

```
(hari@vbox)-[~]  
$ nmap -p 80 localhost  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-05 00:47 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0015s latency).  
Other addresses for localhost (not scanned): ::1  
  
PORT      STATE SERVICE  
80/tcp    open  http  
  
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds
```

Test blocked port:

```
(hari@vbox)-[~]  
$ nmap -p 21 localhost  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-02-05 00:44 EST  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0018s latency).  
Other addresses for localhost (not scanned): ::1  
  
PORT      STATE SERVICE  
21/tcp    closed ftp  
  
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

This proves firewall works.

7) Enable Firewall Logging:

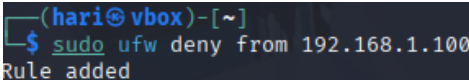
sudo ufw logging on:

```
(hari@vbox)-[~]  
$ sudo ufw logging on  
Logging enabled
```

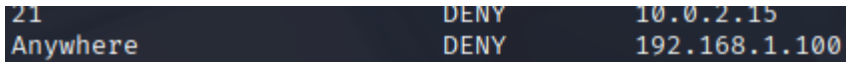
Check logs:

```
(hari@vbox)-[~]  
$ sudo journalctl -k | grep UFW | tail  
Jan 29 06:48:08 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=10.0.2.15 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=24734 DF PROTO=UDP SPT=38202 DPT=3702 LEN=615  
Jan 29 06:48:08 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fe80:0000:0000:0000:0a00:27ff:fe94:4af1 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=566842 PROTO=UDP SPT=55424 DPT=3702 LEN=615  
Jan 29 08:17:21 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fd17:625c:f037:0002:0a00:27ff:fe94:4af1 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=495917 PROTO=UDP SPT=50654 DPT=3702 LEN=615  
Jan 29 08:17:21 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fd17:625c:f037:0002:fe7e:2259:3566:0eba DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=543967 PROTO=UDP SPT=51306 DPT=3702 LEN=615  
Jan 29 08:17:21 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=10.0.2.15 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=45730 DF PROTO=UDP SPT=54333 DPT=3702 LEN=615  
Jan 29 08:17:21 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fe80:0000:0000:0000:0a00:27ff:fe94:4af1 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=967590 PROTO=UDP SPT=36139 DPT=3702 LEN=615  
Jan 29 12:05:31 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fd17:625c:f037:0002:0a00:27ff:fe94:4af1 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=1033773 PROTO=UDP SPT=60990 DPT=3702 LEN=615  
Jan 29 12:05:31 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fd17:625c:f037:0002:fe7e:2259:3566:0eba DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=442402 PROTO=UDP SPT=57667 DPT=3702 LEN=615  
Jan 29 12:05:31 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=10.0.2.15 DST=239.255.255.250 LEN=635 TOS=0x00 PREC=0x00 TTL=1 ID=14595 DF PROTO=UDP SPT=36588 DPT=3702 LEN=615  
Jan 29 12:05:31 vbox kernel: [UFW BLOCK] IN=eth0 OUT= MAC= SRC=fe80:0000:0000:0000:0a00:27ff:fe94:4af1 DST=ff02:0000:0000:0000:0000:0000:0000:000c LEN=655 TC=0 HOPLIMIT=1 FLOWLBL=304111 PROTO=UDP SPT=60790 DPT=3702 LEN=615
```

8) Block a Malicious IP:

Use a test IP: 

Check:



```
21 DENY 10.0.2.15
Anywhere DENY 192.168.1.100
```

IP-based blocking can prevent repeated malicious attempts.

9) Document Firewall Rules:

Rule	Action	Purpose
Default incoming	Deny	Block unauthorized access
Port 22	Allow	SSH management
Port 80	Allow	Web service
Port 21	Deny	Block unused FTP
IP block	Deny	Prevent malicious traffic

10) Explain Impact:

- Reduced attack surface
- Prevented unauthorized access
- Controlled service exposure
- Improved visibility via logs
- Enhanced system security posture